Put $x = 10$

$$y(10) = f(10) = \frac{4(1)(-1)}{(-1)(-4)(-6)}(12) + \frac{(5)(1)(-1)}{(1)(-3)(-5)}$$

$$(13) + \frac{5(4)(-1)}{4(3)(-2)}(14) + \frac{(5)(4)(1)}{6(5)(2)}(16)$$

$$= \frac{1}{6}(12) - \frac{13}{3} + \frac{5(14)}{3\times 2} + \frac{4\times 16}{12}$$

$$= 14.6663$$

Exercise 5.2

1. Using graphic method, find the value of y when $x = 48$ from the following data:

x	40	50	60	70
y	6.2	7.2	9.1	12

2. The following data relates to indirect labour expenses and the level of output

Months	Jan	Feb	Mar
Units of output	200	300	400
Indirect labour expenses (Rs)	2500	2800	3100

Months	Apr	May	June
Units of output	640	540	580
Indirect labour expenses (Rs)	3820	3220	3640

Estimate the expenses at a level of output of 350 units, by using graphic method.

3. Using Newton's forward interpolation formula find the cubic polynomial.

x	0	1	2	3
$f(x)$	1	2	1	10

4. The population of a city in a censes taken once in 10 years is given below. Estimate the population in the year 1955.

Year	1951	1961	1971	1981
Population in lakhs	35	42	58	84

5. In an examination the number of candidates who secured marks between certain interval were as follows:

Marks	0-19	20-39	40-59	60-79	80-99
No. of candidates	41	62	65	50	17

Estimate the number of candidates whose marks are lessthan 70.

6. Find the value of $f(x)$ when $x = 32$ from the following table:

x	30	35	40	45	50
$f(x)$	15.9	14.9	14.1	13.3	12.5

7. The following data gives the melting point of a alloy of lead and zinc where 't' is the temperature in degree c and P is the percentage of lead in the alloy.

P	40	50	60	70	80	90
T	180	204	226	250	276	304

Find the melting point of the alloy containing 84 percent lead.

8. Find $f(2.8)$ from the following table:

x	0	1	2	3
$f(x)$	1	2	11	34

9. Using interpolation estimate the output of a factory in 1986 from the following data.

Year	1974	1978	1982	1990
Output in 1000 tones	25	60	80	170

10. Use Lagrange's formula and estimate from the following data the number of workers getting income not exceeding Rs. 26 per month.

Income not exceeding (₹)	15	25	30	35
No. of workers	36	40	45	48

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